Podcast Background Guide

Using a green screen and OBS with Riverside

Riverside's built-in virtual backgrounds need a computer with at least 8 processor cores, and your laptop (Intel Core i3-1005G1) has 2. That's why you see **"Visual effects unavailable"**. There's nothing wrong with your laptop. It just can't run that particular feature.

The workaround: a free program called **OBS Studio** puts the background in instead. OBS creates a "pretend" webcam called **OBS Virtual Camera**, and you pick that camera in Riverside. With a green screen behind you, this is light enough for your laptop to handle.

> **Setup takes about 20 minutes, once.** After that, each recording session needs just 3 quick steps (see "Every time you record" near the end).

What you need

- **A green screen:** a collapsible one or a smooth green cloth (about $20-40 online). Set it up a few feet behind your chair.
- **Good lighting:** light the green screen evenly, and light yourself separately from the front. Shadows and wrinkles on the screen are the main cause of flickery edges.
- **A background picture** you like (ideally at least 1280 x 720 pixels).
- **Laptop plugged in**, with Windows set to full power: **Settings > System > Power & battery > Power mode > Best performance**.

Step 1: Install OBS Studio

1. Go to **obsproject.com**, download the Windows version and install it.
2. When OBS opens, if the **Auto-Configuration Wizard** pops up, click **Cancel**. You'll set things up by hand in the next step.

Step 2: Set OBS to light settings

In OBS, click **File > Settings**.

- Click **Video** on the left. Set **Base (Canvas) Resolution** and **Output (Scaled) Resolution** both to **1280x720**. Set **Common FPS Values** to **30**.
- Click **Advanced** on the left. Set **Process Priority** to **Above Normal**.
- Click **OK**.

Step 3: Add your webcam

1. In the **Sources** box near the bottom, click **+**, choose **Video Capture Device**, then **OK**.
2. Under **Device**, choose your webcam. You should see yourself.
3. Set **Resolution/FPS Type** to **Custom**, **Resolution** to **1280x720**, and **FPS** to **30**.
4. If there's a **Video Format** option, try **MJPEG**. It's often smoother on built-in webcams.
5. Click **OK**.

Step 4: Remove the green

1. Right-click the webcam in the Sources list and choose **Filters**.
2. Under **Effect Filters**, click **+** and choose **Chroma Key**, then **OK**.
3. Set **Key Color Type** to **Green**, then use these starting values:

Setting	Start at	Adjust if...
Similarity	400	Green still shows: raise it. Parts of you vanish: lower it.
Smoothness	80	Edges look harsh or jagged: raise it a little.
Key Color Spill Reduction	100	Green tint on hair or edges: raise it.

4. Close the Filters window.

Step 5: Add your background picture

1. In **Sources**, click **+**, choose **Image**, then **OK**. Browse to your picture and click **OK**.
2. Right-click the picture and choose **Transform > Fit to screen**.
3. In the Sources list, drag the picture so it's **below** your webcam. (Things lower in the list appear behind things higher up.)

You should now see yourself in front of your background.

Step 6: Turn on the pretend webcam

1. Click **Start Virtual Camera** (bottom right, in the Controls area).
2. Optional, but it helps your laptop: right-click the big preview picture and untick **Enable Preview**. OBS keeps working without displaying it.

Step 7: Pick it in Riverside

1. Open Riverside in **Google Chrome** and go into your studio.
2. In the camera settings, choose **OBS Virtual Camera** as your camera.
3. Keep Riverside's own background and blur effects **off**. OBS is doing that job now.
4. If the studio has a video quality setting, choose **720p**.

Step 8: Do a test recording

- Record 2-3 minutes and play it back. Look for stuttering video, audio out of sync, or flickering edges.
- During the test, open **Task Manager** (press **Ctrl + Shift + Esc**) and look at **CPU**. If it sits near 100%, close other programs and extra Chrome tabs.
- Still near 100%? In Steps 2 and 3, change **30** FPS to **24**.

Every time you record

OBS remembers all your settings. Just do this:

1. Open **OBS**.
2. Click **Start Virtual Camera**.
3. Open **Riverside** in Chrome and choose **OBS Virtual Camera**.

Leave OBS open until you've finished recording. Closing it turns off your camera in Riverside.

If something goes wrong

Problem	Try this
Riverside doesn't list **OBS Virtual Camera**	Close Chrome completely. In OBS, click Start Virtual Camera first, then reopen Chrome.
The picture is black	Another app (Zoom, Teams, the Camera app) may be using the webcam. Close it, then restart OBS.
Patches of green still show	Improve the lighting on the green screen and smooth out wrinkles first. Then raise Similarity a little.
Edges of hair or shoulders flicker	Light the screen more evenly, sit a bit further in front of it, and raise Smoothness slightly.
Video is choppy or laggy	Plug in the laptop, close other programs and tabs, untick Enable Preview, or drop to 24 FPS.
You look green-tinted	Raise Key Color Spill Reduction, or move a little further from the green screen.